

G-CAF

GovDigital Composable Architecture Framework

Volume 12

Composition & UI Extension

v2.0

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Chapter 1

Composition Overview

Composition Contracts define interface standards that PBCs must implement to be composable within experience platforms at the Experience Tier.

Problem Statement

Without Contracts	Impact
Custom integration per PBC	High cost, slow time-to-compose
Inconsistent user experience	Users learn different patterns per capability
Tight coupling	Platform changes break PBCs and vice versa
Channel-specific code	Same capability reimplemented per channel
Siloed search and tasks	Users check multiple places for information

Solution — 7 Composition Contracts

Contract	Module	Purpose
UI Contribution	G-CAF-075a	Expose UI fragments (widgets, panels, pages) to platforms
Task Contribution	G-CAF-075b	Surface actionable items to unified inbox
Search Contribution	G-CAF-075c	Enable federated cross-PBC search
Navigation Contribution	G-CAF-075d	Support deep linking and dynamic routing
Notification Contribution	G-CAF-075e	Publish to unified notification center
Context Acceptance	G-CAF-075f	Receive and use composition context
Accessibility Contribution	G-CAF-075g	Ensure inclusive access (WCAG, RTL)

Chapter 2

Design Principles

Principle	Description	Rationale
Declarative over Imperative	PBCs declare capabilities; platforms consume capabilities	Loose coupling, independent evolution
Event-Driven Federation	Real-time events over polling for tasks, notifications	Scalability, freshness
Schema-First	All contracts defined as versioned YAML/JSON	On-specification checking, documentation
Progressive Enhancement	Basic composition works; advanced features optional	Incremental adoption possible
Channel Agnostic	Contracts don't assume specific channels	Omni-channel support (web, mobile, kiosk, chatbot)

NS Compliance

NS Theorem	Application
Separation of Concerns	Each contract has a single change driver
Data Version Transparency	All schemas follow additive-only evolution
Action Version Transparency	Contract versions independent of PBC versions
Separation of State	Composition platforms don't store PBC state

Schema Evolution Rules

Allowed	Not Allowed
Add optional field	Remove field
Add new enum value	Rename field
Add new endpoint	Change field type
Extend with new capability	Change required to optional
Deprecate with replacement	Remove without deprecation

Chapter 3

UI Contribution Contract

Enables PBCs to contribute UI components to composition platforms without platform-specific integration code.

Contribution Types

Type	Description	Use Case
Widget	Small, embeddable UI component	Dashboard cards, summary panels
Panel	Medium-sized content area	Detail views, forms
Page	Full-page experience	Complete workflows
Modal	Overlay dialog	Quick actions, confirmations
Inline	Embedded within host content	Rich previews, expansions

Manifest

PBCs publish a UI manifest at `/well-known/composition/ui-manifest.yaml` declaring all UI contributions with type, mount point, dimensions, required context, supported channels, and lazy-loading configuration.

Interfaces: UI Manifest (REST), Widget Renderer (Web Components/iFrame), State Bridge (JavaScript), Theme Provider (CSS/JS). Error handling covers widget load failures (show fallback with retry), theme mismatches (apply default), and invalid manifests (return validation errors).

Chapter 4

Task Contribution Contract

Surfaces actionable items to a unified inbox — approvals, tasks, notifications, reminders, and assignments.

Task Types

Type	Description	User Action
Approval	Requires decision (approve/reject)	Binary or multi-option decision
Action	Requires user to do something	Navigate and complete
Notification	Informational, may need acknowledgment	Read, dismiss, or act
Reminder	Time-based prompt	Snooze, complete, or dismiss
Assignment	Work item assigned to user	Accept, delegate, or complete

Task Schema (Key Fields)

task_id (UUID), task_type, source_pbc, title (localized en/ar), description, priority (critical/high/medium/low), created_at, due_at, expires_at, assignee (employee/role/group/dynamic), available_actions. Events: task.created, task.updated, task.completed, task.expired.

Chapter 5

Search Contribution Contract

Provides federated search across all composed PBCs — a single search experience with unified results from multiple sources.

Search Contribution Types

Type	Description	Example
Entity Search	Search business entities	Employees, cases, requests
Document Search	Search documents/files	Policies, forms, attachments
Content Search	Search textual content	Knowledge base, news articles
Action Search	Search available actions	"Submit leave", "View payslip"

Manifest published at `/.well-known/composition/search-manifest.yaml` declaring searchable fields with boost factors, filterable facets, result display templates, and supported query types. Results returned in unified format with relevance scoring.

Chapter 6

Navigation Contribution Contract

Enables dynamic navigation, deep linking, and consistent cross-PBC navigation experience.

Navigation Components

Component	Description	Example
Routes	Navigable destinations with URL patterns	/leave, /leave/request/new
Breadcrumbs	Navigation path indicators	Home > Leave > New Request
Quick Actions	Command palette entries for rapid access	"Request Leave", "View Payslip"
Context Links	Situational navigation suggestions	"View Leave History" on employee profile

Manifest at **`/well-known/composition/navigation-manifest.yaml`** declares routes (with parameters), breadcrumb providers, quick actions with keywords, and deep link resolution endpoints.

Chapter 7

Notification Contribution Contract

Publishes notifications to a unified center with multi-channel delivery.

Notification Types & Channels

Type	Visual	Channels Available
Info	Blue indicator	In-app, push, email
Success	Green indicator	In-app, push
Warning	Yellow indicator	In-app, push, email, SMS
Error	Red indicator	In-app, push, email, SMS
Action Required	Orange + badge	All channels including WhatsApp

Notification schema includes: notification_id (UUID), type, source_pbc, correlation_id (group related), title/body (localized en/ar), target (employee/role/group/broadcast), channels config, priority, deep_link, expires_at, and available actions. User preferences control which channels are used.

Chapter 8

Context Acceptance Contract

Standardizes how platforms pass context to PBCs for seamless cross-application experience.

Context Categories

Category	Description	Key Fields
User	Authenticated user info	employee_id, roles, entity_id, locale, timezone, display_name (en/ar)
Session	Current session details	session_id, channel (web/mobile/kiosk), device type
Navigation	Current routing state	current_route, route_params, referrer
Features	Feature flags and settings	dark_mode, experimental_features, a11y preferences
Custom	Platform-specific extensions	Platform-defined additional context

Context is delivered via JavaScript API (for embedded widgets), REST API (for server-side), and event-driven updates (for real-time changes like role switches or locale changes).

Chapter 9

Accessibility Contribution Contract

Ensures all composed PBCs meet accessibility standards for inclusive government services.

Core Requirements

Requirement	Specification	WCAG Reference
Color Contrast	4.5:1 minimum for text; 3:1 for large text	1.4.3
Focus Indicators	Visible focus states on all interactive elements	2.4.7
Keyboard Navigation	All functions accessible via keyboard alone	2.1.1
Error Identification	Clear, descriptive error messages	3.3.1
Labels	All form inputs have associated labels	1.3.1
Alt Text	All images have descriptive alt text	1.1.1
Resize	Content readable at 200% zoom	1.4.4
Motion	Respect prefers-reduced-motion setting	2.3.3

Compliance Standards

Standard	Requirement
WCAG	2.1 Level AA minimum for all public-facing experiences
Screen Reader	Full ARIA compatibility; all interactive elements properly labeled
RTL Support	Full Arabic right-to-left layout support
Keyboard	Complete keyboard access to all functions

PBCs must publish an accessibility manifest declaring compliance level, audit results, and user preference support. Automated accessibility audits must pass before Go-Live Gateway approval.

Chapter 10

Platform Requirements

Responsibilities that composition platforms (TAMM, OneHub) must provide to consume contracts.

Responsibility	Description
Manifest Discovery	Discover and cache PBC manifests at /.well-known/composition/ (default 5-min cache)
Widget Orchestration	Lazy load, initialize, manage lifecycle; widget failures don't crash platform
Context Management	Build context from auth/session/navigation; propagate to all mounted widgets
Event Routing	Route events between PBCs and platform; bi-directional message passing
Search Federation	Aggregate search across PBCs; merge and rank results; handle timeouts with partial results
Task Aggregation	Consolidate tasks into unified inbox; support bulk operations
Notification Delivery	Route notifications to appropriate channels based on user preferences
Navigation Management	Build dynamic navigation from manifests; handle deep links and breadcrumbs
Error Handling	Graceful degradation when individual PBCs fail; show fallbacks, not errors

Chapter 11

Governance & Implementation

Contract Lifecycle

State	Description	Transition Approval
Draft	Under development; any changes allowed	Contract owner
Review	DDA review; minor changes only	DDA
Approved	Ready for implementation; no changes	DDA
Active	In production use; patch only	Go-Live Gateway
Deprecated	Being phased out; no changes	DDA (MINOR: 6mo) / ARB (MAJOR: 12mo)
Retired	No longer supported	DDA (zero consumers verified)

Version Management

Change Type	Version Impact	Approval	Deprecation Window
Breaking change	MAJOR increment	ARB	12 months
New capability	MINOR increment	DDA	N/A (backward compatible)
Bug fix	PATCH increment	DDA	N/A

3-Level Adoption Guide

Level	Contracts	Suitable For	Time Estimate
Level 1: Basic	Navigation + Context	Simple PBCs, API-only, legacy wrappers	2-5 days
Level 2: Standard	+ UI + Task + Notification + Basic Search	Most PBCs	1-2 weeks
Level 3: Advanced	+ Rich widgets + Full search + Personalization	Complex, high-engagement PBCs	2-4 weeks